

SentinellQ - Ensemble Anomaly Detection Report

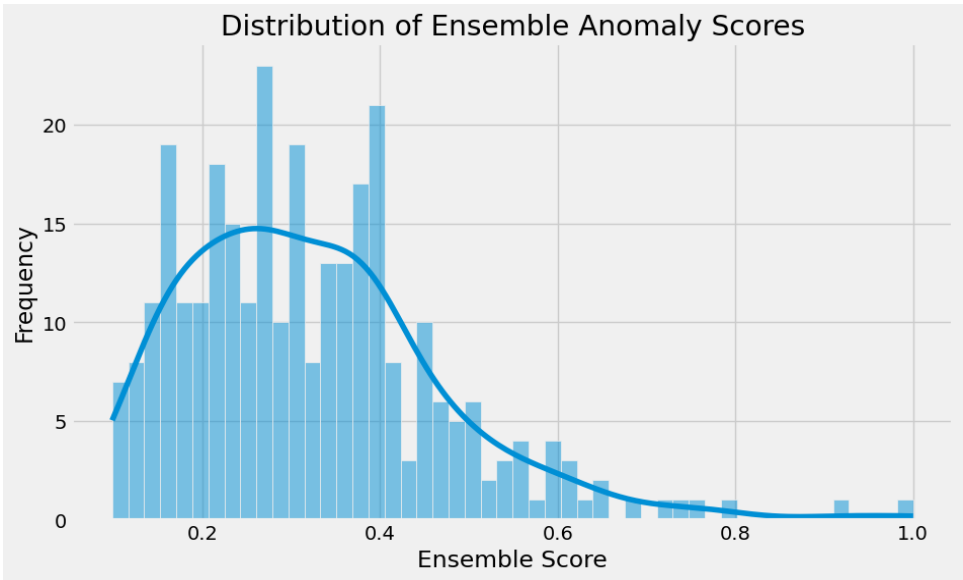
Executive Summary

This report details the performance and prioritization of the final ensemble anomaly detection model. The ensemble combines Isolation Forest, One-Class SVM, EWMA, CUSUM, Persistence Score, and Incident Probability to provide a robust anomaly detection signal. Anomalies are prioritized based on their ensemble scores.

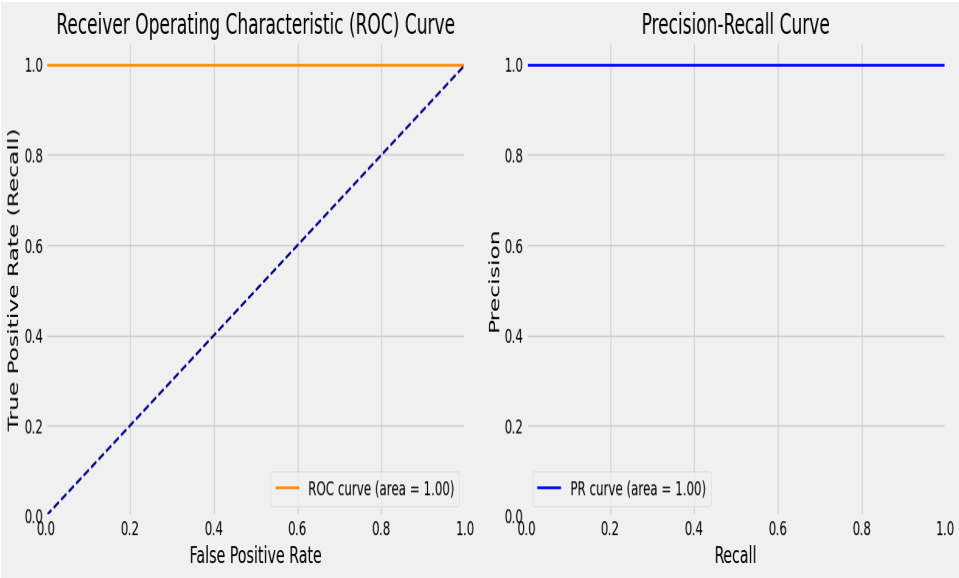
Ensemble Model Evaluation

Metric	Value
Model	Final Ensemble
Precision	1.0000
Recall	0.0579
F1	0.1094
ROC_AUC	0.9993
PR_AUC	0.9887

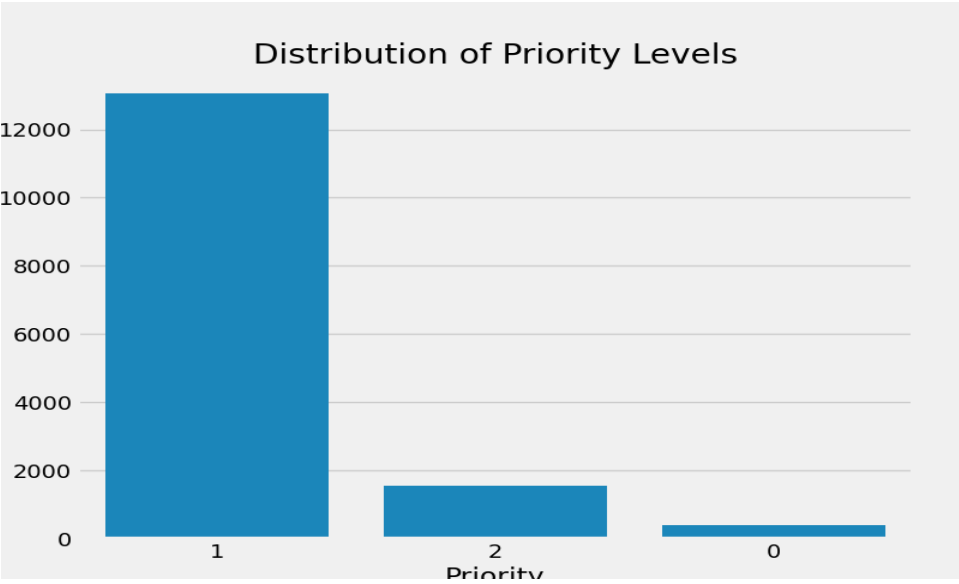
Distribution of Final Ensemble Anomaly Scores



ROC and Precision-Recall Curves



Distribution of Anomaly Priority Levels



Top 10 Highest Scoring Anomalies

Timestamp	Service	Latency	Score	Priority
2026-01-01 14:26:00	payment-api	243	0.8182183772912701	Critical
2026-01-01 13:23:00	notification-service	274	0.8109358464302903	Critical
2026-01-01 14:50:30	portfolio-service	170	0.8099581648847477	Critical
2026-01-04 16:45:30	portfolio-service	241	0.8028005434681588	Critical

2026-01-01 13:36:30	fraud-detection	250	0.8001260661628544	Critical
2026-01-04 15:56:30	trading-engine	291	0.7966756813228537	Critical
2026-01-01 13:35:30	portfolio-service	269	0.7954106623394366	Critical
2026-01-01 14:18:30	fraud-detection	250	0.787764298713662	Critical
2026-01-04 17:10:30	payment-api	231	0.7763846623125074	Critical
2026-01-01 14:19:30	trading-engine	150	0.7734712490486381	Critical